



TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC2 2021

H1 BIOLOGY

Paper 1 Multiple Choice

8876/01

16 September 2021

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the Multiple Choice Answer Sheet (MCAS).

Write in soft pencil.

Do **NOT** use staples, paper clips, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

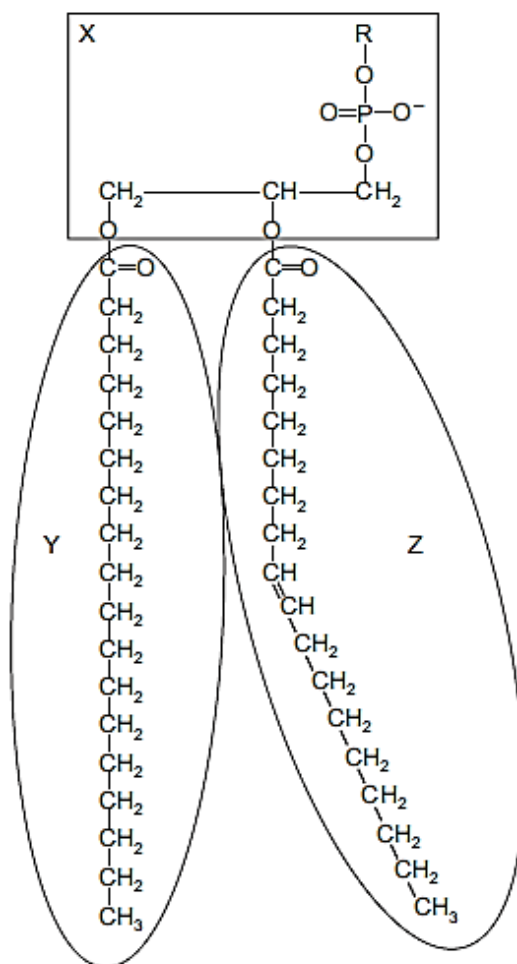
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

Answer **all** the questions.

- 1 The diagram shows a phospholipid molecule divided into three regions, X, Y and Z. R, in region X, represents a range of possible chemical groups.

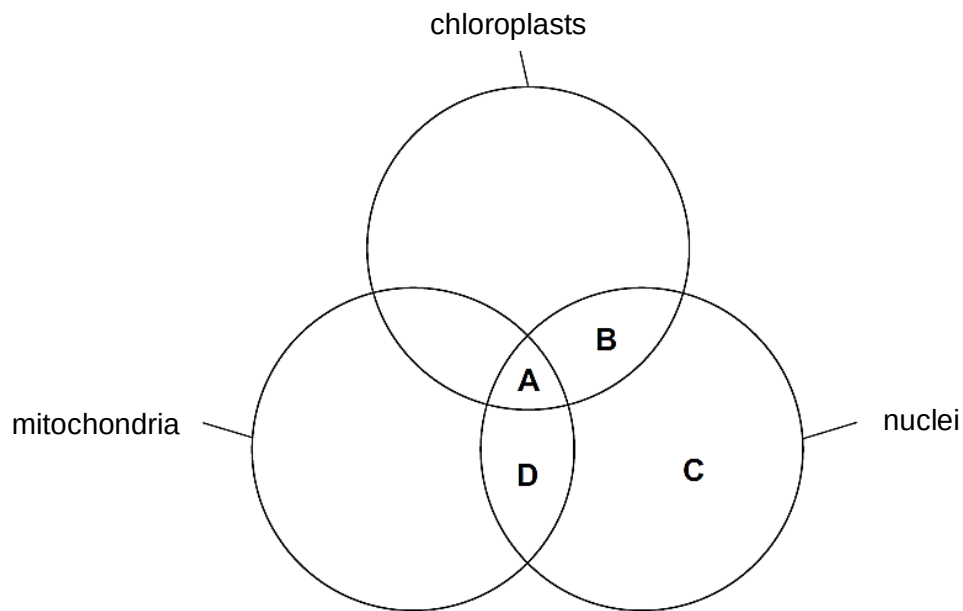


Regions X, Y and Z affect the properties of cell surface membranes in different ways.

Which row shows the effect of each region on the properties of a cell surface membrane?

	increases fluidity of hydrophobic region	repels polar molecules	attracts water molecules
A	X	X	Y and Z
B	Y	Y and Z	X
C	Y and Z	X	Y and Z
D	Z	Y and Z	X

- 2 Which letter identifies cell structures where semi-conservative replication of DNA occurs?



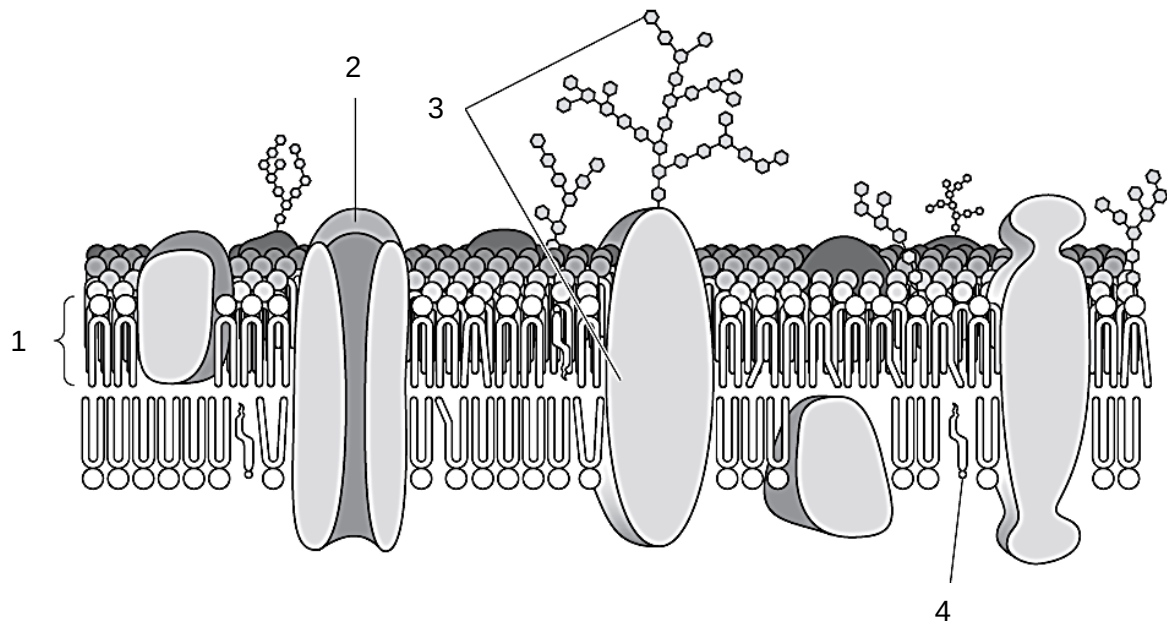
- 3 Some cell structures are listed in a particular order.

- 1 nucleus
- 2 ribosome
- 3 Golgi body
- 4 vesicle

What determines the order in which these cell structures are listed?

- A sequence used in synthesis of a lipid
- B sequence used in synthesis of a glycoprotein
- C size from largest to smallest
- D size from smallest to largest

4 The diagram shows a cell surface membrane.



Which statements about the labelled molecules in the membrane are correct?

- 1 is involved in the diffusion of ions.
- 2 is involved in facilitated diffusion.
- 3 is involved in the recognition of foreign substances.
- 4 is involved in membrane fluidity.

- A** 1 and 3
- B** 1 and 4
- C** 2 and 4
- D** 1, 2 and 3

5 Which statements about the differences between phospholipids and triglycerides are correct?

- 1 Phospholipids have hydrophobic regions but triglycerides do not.
- 2 The fatty acids in a phospholipid are always saturated but in a triglyceride they may be saturated or unsaturated.
- 3 Phospholipids are polar molecules but triglycerides are non-polar.

A 1

B 3

C 1 and 2

D 2 and 3

6 An investigation was carried out into the effect of different treatments on the permeability of the cell surface membranes and tonoplasts (membranes surrounding vacuoles) of beetroot cells. Beetroot cell vacuoles contain a red pigment. This pigment is unable to pass out of the cells because it cannot diffuse through the tonoplasts or cell surface membranes.

1 cm³ cubes were cut from beetroot tissue and washed in running water for 20 minutes to remove any pigment released from damaged cells.

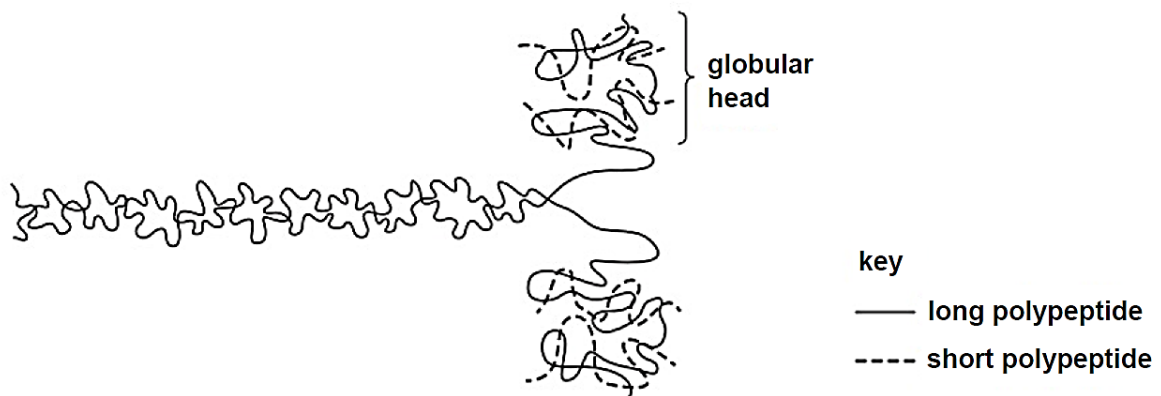
The cubes were then placed in test-tubes with different contents and observed for five minutes.

Which row shows a correct explanation for the observation recorded for one of the treatments?

	treatment	observation	explanation
A	dilute hydrochloric acid	contents of test-tube remain clear	membrane proteins have been denatured
B	ethanol	contents of test-tube remain red	lipids, including membrane phospholipids, have dissolved
C	water at 20°C	contents of test-tube remain clear	membrane proteins have been denatured
D	water at 20°C	contents of test-tube remain red	lipids, including membrane phospholipids, have dissolved

[Turn over

7 The diagram shows a protein molecule.



Two long polypeptides each form α -helices for much of their length and these twist together into a fibre.

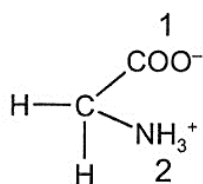
At one end, each of these polypeptides coils into a globular head.

Two short polypeptides bind to each globular head.

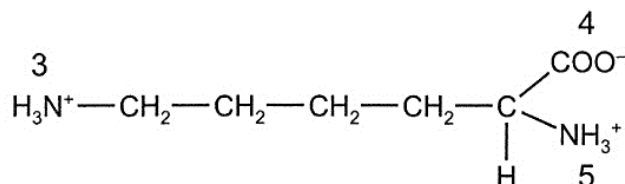
What describes the protein structure of this molecule?

- A** secondary structure because the long polypeptides form α -helices
- B** tertiary structure because the α -helices form a fibre
- C** tertiary structure because the heads form globular proteins
- D** quaternary structure because each molecule consists of six polypeptides

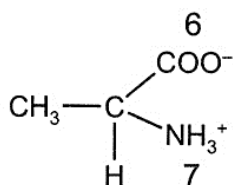
8 The diagrams show the structure of four amino acids.



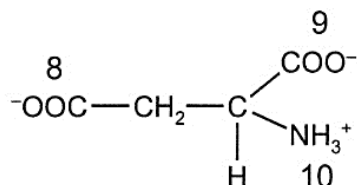
glycine



lysine



alanine



aspartate

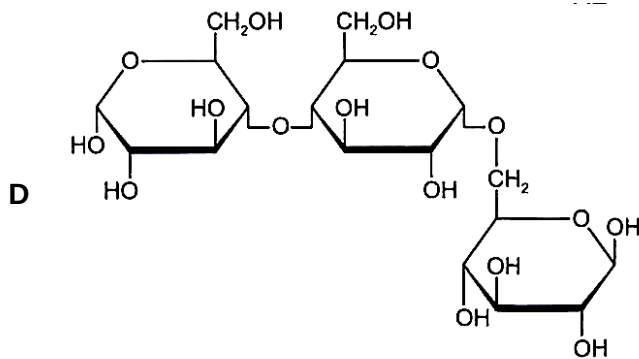
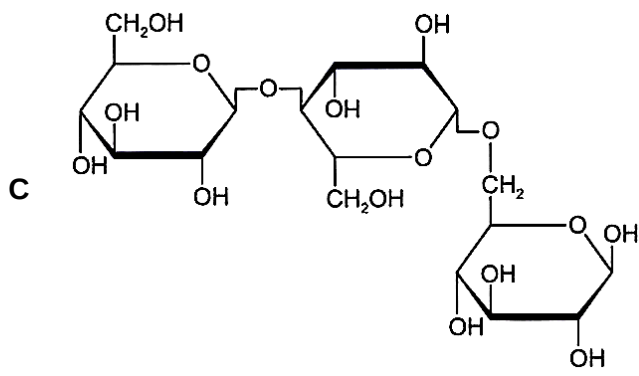
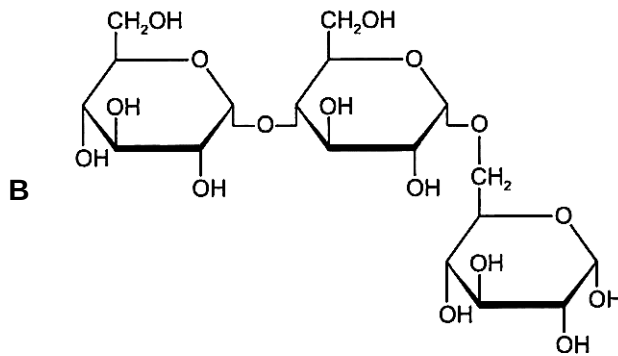
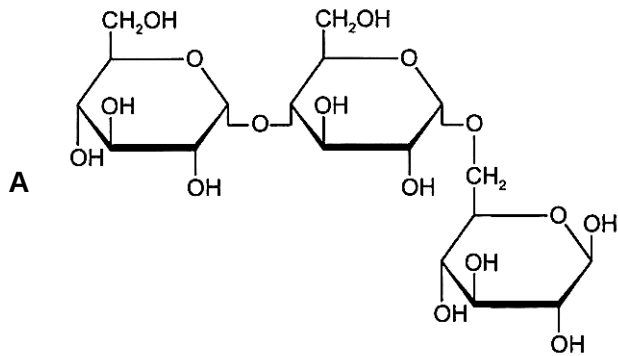
During the synthesis of a peptide, these amino acids are joined in the sequence shown.

amino end carboxyl end
aspartate – alanine – glycine – lysine

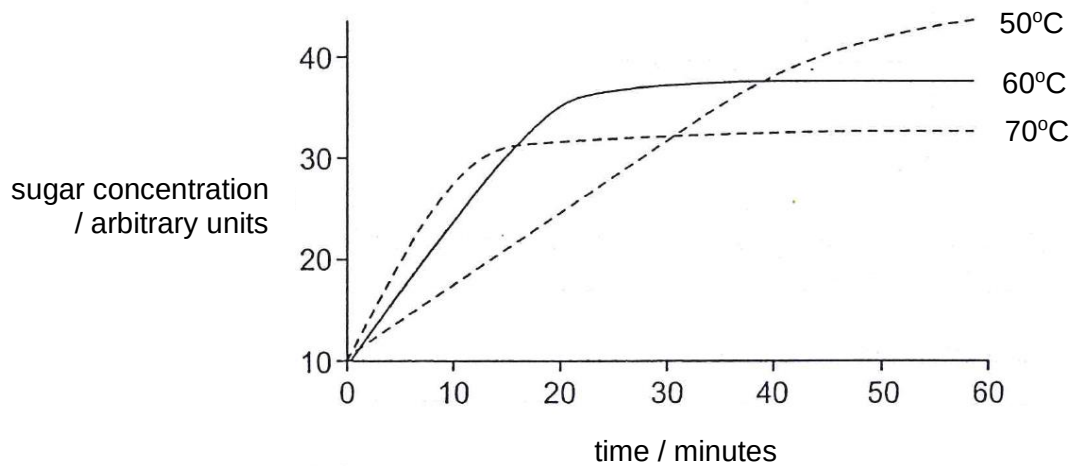
In which order do the numbered groups on each amino acid condense to form the peptide bonds in this peptide?

- A 3 and 1, 2 and 6, 7 and 8
- B 5 and 1, 2 and 6, 7 and 9
- C 8 and 7, 6 and 2, 1 and 3
- D 9 and 7, 6 and 2, 1 and 5

- 9 Which diagram correctly shows a trisaccharide containing both 1,4-glycosidic and 1,6-glycosidic bonds, and formed from both α -glucose and β -glucose molecules?



10 In beer-making, enzymes present in malting barley hydrolyse starch into sugar, ready for fermentation. The graph shows the production of sugar at three different temperatures over a period of 60 minutes. All conditions, including the starting amount of starch, were standardised.



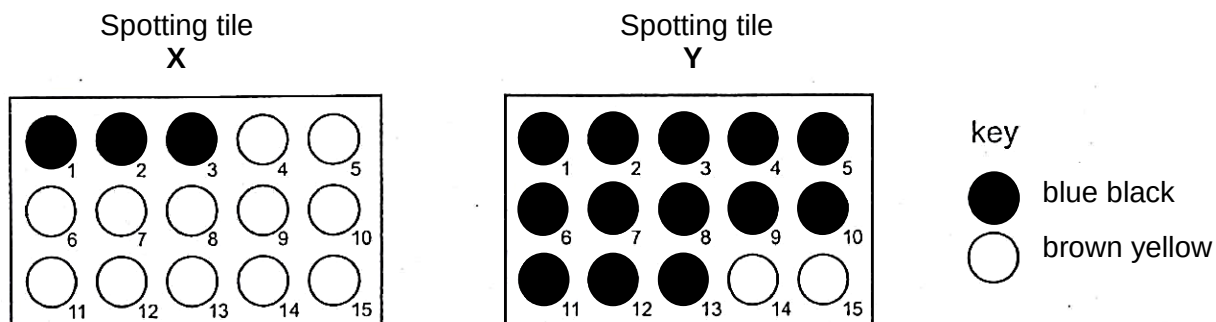
What does the graph show?

- 1 At 60°C, all the starch is hydrolysed within 30 minutes.
- 2 At 70°C, the enzymes are denatured before hydrolysis is complete.
- 3 The difference in the rate of enzyme activity can be calculated in the first 10 minutes.
- 4 The amount of sugar produced from starch is more than 40 arbitrary units.

- A** 1 and 4
- B** 2 and 3
- C** 1, 2 and 4
- D** 2, 3 and 4

- 11** An experiment was carried out to investigate the digestion of starch using amylase at two different temperatures. A sample was removed from each mixture at 15 second intervals and placed onto a spotting tile well containing two drops of iodine in KI solution. Completion of the digestion of starch will result in a brown yellow mixture when iodine is added.

The results are shown in the diagram.

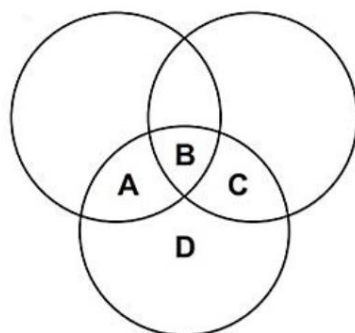


Which shows the correct temperatures and times for the complete digestion of starch?

	spotting tile X		spotting tile Y	
	temperature / °C	time / s	temperature / °C	time / s
A	30	45	10	195
B	30	195	10	45
C	10	45	30	195
D	10	195	30	45

- 12** Which processes involve mitosis?

formation of daughter cells which are not genetically identical



development of a zygote into an adult human being

replacing skin cells damaged by an injury

13 Each of these descriptions of a stage of mitosis in an animal cell is incomplete.

- 1 centromeres separate and move to the poles of the cell
- 2 chromatin condenses, centrioles migrate to the poles of the cell
- 3 chromosomes become less condensed
- 4 sister chromatids joined by centromeres align at the equator of the cell

Which row completes the description of each mitotic stage?

	1	2	3	4
A	nuclear membrane and nucleolus reform	sister chromatids are pulled apart by shortening of spindle microtubules	centromeres bind to spindle microtubules	spindle microtubules appear between centrioles and nuclear membrane disappears
B	centromeres bind to spindle microtubules	nuclear membrane and nucleolus reform	spindle microtubules appear between centrioles and nuclear membrane disappears	sister chromatids are pulled apart by shortening of spindle microtubules
C	spindle microtubules appear between centrioles and nuclear membrane disappears	centromeres bind to spindle microtubules	sister chromatids are pulled apart by shortening of spindle microtubules	nuclear membrane and nucleolus reform
D	sister chromatids are pulled apart by shortening of spindle microtubules	spindle microtubules appear between centrioles and nuclear membrane disappears	nuclear membrane and nucleolus reform	centromeres bind to spindle microtubules

[Turn over

14 Two students were discussing the involvement of DNA and RNA in transcription and translation.

- Student 1 always stated correct facts.
- Student 2 gave further information, which was sometimes correct.

	correct facts given by student 1	further information given by student 2
1	A length of mRNA is 747 nucleotides long, including stop and start codons.	This mRNA can produce a polypeptide that is 249 amino acids long.
2	Adjacent mRNA codons of AAU and CUG bind to complementary tRNA anticodons.	There is a total of 14 hydrogen bonds formed between these two codons and their anticodons.
3	RNA polymerase catalyses the formation of the mRNA from the template strand of DNA.	During translation, an RNA adenine nucleotide will pair with a DNA thymine nucleotide.
4	A DNA adenine nucleotide is structurally different to an RNA adenine nucleotide.	The difference is in the pentose sugars; DNA is deoxyribose and RNA is ribose.

Which further information, given by student 2, is correct?

- A** 1 and 3
- B** 2 and 4
- C** 1, 2 and 4
- D** 2, 3 and 4

- 15** A short piece of DNA, 18 base pairs long, was analysed to find the number of nucleotide bases in each of the polynucleotide strands.

Some of the results are shown in the table.

	number of nucleotide bases			
	adenine	cytosine	guanine	thymine
strand 1		4		7
strand 2		5		

How many nucleotides containing thymine were present in strand 2?

- A** 2
- B** 4
- C** 5
- D** 7

- 16** In order for DNA to function as an effective data storage molecule in cells, it must be:

- able to carry information
- stable
- accurately copied to daughter cells.

The following are all features of DNA structure.

- 1 hydrogen bonds between bases
- 2 covalently bonded sugar-phosphate backbone
- 3 complementary base pairing
- 4 specific sequences of bases.

Which features of DNA structure, from 1, 2, 3 and 4, are important to each aspect of its function as an effective data storage molecule?

	carries information	stable	accurately copied to daughter cells
A	1 and 2	2 and 4	1 and 4
B	3	1 and 3	2
C	1 and 4	3 and 4	1, 2 and 3
D	4	1 and 2	3

[Turn over

- 17** Huntington's disease (HD) is a rare inherited neurological disorder caused by a single dominant allele of a gene. The disease's symptoms are characterised by abnormal body movements and a lack of coordination, but it also affects a number of mental abilities and some aspects of personality. These physical symptoms commonly become noticeable in a person's forties.

Which of the following is possibly true about the inheritance of HD?

- A** As long as one of the parents has HD, all the offspring will have HD.
 - B** No offspring of two normal individuals will have HD.
 - C** The phenotypic expression of HD is also dependent on the environment which means that the symptoms only appear when the person has aged to 40 years.
 - D** The dominant allele could be carried on a sex chromosome.
- 18** In the fruit fly *Drosophila melanogaster* the gene for eye colour is sex-linked. The allele for red eye (R) is dominant to the allele for white eye (r).

A cross between two flies produced the offspring shown in the table below.

sex of offspring	number with white eyes	number with red eyes
female	23	22
male	21	22

The genotypes of the parents in this cross were

- A** X^rX^r and X^RY
- B** X^RX^r and X^rY
- C** X^RX^r and X^RY
- D** X^RX^R and X^rY

- 19** Seeds from a pure breeding plant were planted in identical pots of compost and watered regularly. Sets of ten pots were placed in different light conditions and left until the first leaves had developed.

The table shows the mean height for the young stems, mean length of first leaf and the colour of the leaves.

	no light	dim light	bright light
mean height / cm	8	6	4
mean leaf length / cm	1.5	1.4	1.4
colour of leaves	pale yellow	pale green	dark green

Which explains the effect of light on the phenotype of the young plants?

- A** The activity of genes involved in chlorophyll synthesis and stem growth varies with light intensity.
- B** The activity of genes involved in stem and leaf growth is decreased by light.
- C** The genes involved in chlorophyll synthesis and stem growth are activated by light.
- D** The genes involved in chlorophyll synthesis and stem growth are inactivated by light.
- 20** Which of the following give rise to genetic variation in a population?
- 1 Natural selection favouring one phenotype over another phenotype.
 - 2 Spontaneous gene mutations.
 - 3 Formation of gametes.
 - 4 Random chance of gametes fusing with each other.
- A** 1 and 4
- B** 2 and 3
- C** 2, 3 and 4
- D** 1, 2, 3 and 4
- 21** Which of the following explains why population is the smallest unit of evolution?
- A** Allele frequencies in a population change due to natural selection acting on individuals.
- B** Allele frequencies of individuals change due to the effect of natural selection.
- C** Natural selection acts on populations.
- D** Populations adapting to selection pressures.

[Turn over

22 Which of the following results in the transfer of electrons down the electron transport chains during the light dependent reactions of photosynthesis?

- A** NADP is converted to NADPH.
- B** Water is split by photolysis.
- C** ATP is synthesised.
- D** Pigment molecules absorb energy.

23 The following substances are products of fermentation.

- 1 ATP
- 2 lactate
- 3 carbon dioxide

Which of these are products of fermentation in human muscle cells?

- A** 2 only
- B** 1 and 2
- C** 2 and 3
- D** 1, 2 and 3

24 In 2016, the United Kingdom government allowed scientists to genetically modify cells obtained from very early human embryos.

Which statement correctly describes an ethical concern about this research?

- A** Genetically modified iPSCs could give rise to genetically modified individuals.
- B** Genetically modified multipotent stem cells could give rise to genetically modified individuals.
- C** Genetically modified pluripotent stem cells could give rise to genetically modified individuals.
- D** Genetically modified totipotent stem cells could give rise to genetically modified individuals.

25 The table shows the DNA triplet codes for some amino acids.

amino acid	DNA triplet code	amino acid	DNA triplet code
arginine	GCA	glycine	CCA
arginine	GCC	glycine	CCG
arginine	GCG	glycine	CCT
asparagine	TTA	lysine	TTC
asparagine	TTG	lysine	TTT
cysteine	ACA	proline	GGA
cysteine	ACG	proline	GGC
STOP	ATC	valine	CAC

The base sequence on the DNA template strand coding for part of a polypeptide is shown.

CCA TTC ACG GCG TTA GCA

Two mutations occur in this sequence during DNA replication.

Which mutated DNA would result in a polypeptide with one different amino acid?

- A** CCA ATC ACG GCG TTG GCA
- B** CCA TTC ACA GCA TTA GCA
- C** CCA TTC ACG CCG TTA GCC
- D** CCT TTC ACG GCG TTA GCC

- 26** One individual, organism P, in a sexually reproducing population of mammals has two forms of a protein in its heart muscle. The normal form is identical to that in most other members of the population, while the variant form differs by one amino acid. The diagram shows the amino acid sequences in part of the normal protein and in the same part of the variant protein.

normal protein
-glu-lys-lys-gly-glu-

variant protein
glu-lys-gly-gly-glu

What could account for the presence of the normal protein and the variant protein in P?

- 1 The sequence of amino acids was altered in a cell in the heart muscle of P's mother.
- 2 The sequence of amino acids was altered in a cell in the ovaries of P's mother.
- 3 The sequence of bases was altered in the DNA in a cell in the heart muscle of P.
- 4 The sequence of bases was altered in the DNA in a cell in an ovary of P.
- 5 The sequence of bases was altered in the DNA in a cell in a testis of P's grandfather.

- A** 1 and 3
B 3 and 5
C 1, 2 and 4
D 2, 4 and 5

- 27** Sickle cell anaemia is caused by a mutation in an allele of the gene that codes for the β -globin polypeptide of haemoglobin.

The diagram shows the sequence of bases in a small section of the template strand of DNA for both the Hb^A (normal) and Hb^S (sickle cell) β -globin alleles.

Hb ^A	CTGACTCCTGAGGAGAAGTCT
Hb ^S	CTGACTCCTGTGGAGAAGTCT

How will the mutation in the allele result in the production of an altered version of the β -globin polypeptide?

- A** A tRNA molecule with the anticodon GUG will form hydrogen bonds with the altered codon on mRNA.
B All the amino acids coded for after the mutation will differ from those in the Hb^A protein.
C mRNA transcribed from the Hb^S allele will contain the codon CAC instead of the codon CTC.
D The ribosome will be unable to continue translation of the Hb^S mRNA after the altered codon.

- 28** About 20% of all human pregnancies are estimated to be lost by spontaneous abortion (miscarriage). About half of these spontaneous abortions are associated with chromosomal aberrations.

The table shows the percentage incidence of each type of chromosome aberration that was detected in spontaneous abortions associated with chromosome aberrations.

chromosome aberration	percentage incidence
trisomy ($2n + 1$)	62
monosomy ($2n - 1$)	18
triploidy ($3n$)	17
translocations	3

What may be concluded from these data?

- 1 Numerical aberrations are much more likely to be associated with spontaneous abortions than are structural aberrations.
 - 2 The occurrence of three sets of chromosomes has a similar percentage incidence as monosomy.
 - 3 The presence of an extra chromosome is more likely to be associated with spontaneous abortion than is the lack of one chromosome.
- A** 1 and 2
- B** 1 and 3
- C** 2 and 3
- D** 1, 2 and 3

29 Leukaemia is a cancer of blood stem cells.

People in whom the *JAK2* gene mutates before the *TET2* gene, show symptoms of pre-leukemia (an early form of the disease) 10 years earlier than those in whom the *TET2* gene mutates first.

Between 5% and 10% of people with this pre-leukaemia go on to develop leukaemia.

People in whom the *TET2* gene mutates before the *JAK2* gene are more likely to develop leukaemia than people in whom the *JAK2* gene mutates first.

What is supported by these findings?

- 1 Leukaemia may develop as a result of mutations in DNA.
- 2 Mutations of the *JAK2* gene and the *TET2* gene are each involved in the development of leukaemia.
- 3 The order in which mutations occur affects the development of leukaemia.
- 4 People with mutations of the *JAK2* gene develop leukaemia earlier than people with mutations of the *TET2* gene.

A 3 and 4

B 1, 2 and 3

C 1, 2 and 4

D 1, 2, 3 and 4

30 Human activity is believed to have contributed to climate change. It has been suggested that climate change has resulted in a worldwide loss of biodiversity.

Which statements explain why there may have been a loss of biodiversity?

- 1 habitat loss
- 2 invasion from immigrating species that replace native species
- 3 climatic conditions change faster than species can adapt by evolution
- 4 colonisation of previously uninhabited areas

A 1 and 3

B 2 and 4

C 1, 2 and 3

D 1, 2 and 4